Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/18/2006
PI Name:	Miller, Jack Ph.D.		
Project Title:	Measurements of Fragmentation Cross Sections and Particle Spectra for Galactic Cosmic Ray-like Nuclei		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	miller@lbl.gov	**Fax:**	FY 510 486-6949
PI Organization Type:	GOVERNMENT	**Phone:**	510 486-7130
Organization Name:	Ernest O. Lawrence Berkeley National Laboratory		
PI Address 1:	Mail Stop 74-197, Life Sciences Division		
PI Address 2:	1 Cyclotron Road		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-05
Start Date:	01/01/2003	**End Date:**	01/01/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	10	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhattacharya, M	**Contact Phone:**	256 961 7546
Contact Email:	bhattmx@nsstc.uah.edu		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zeitlin, C. (Lawrence Berkeley National Laboratory) Heilbronn, L. (Lawrence Berkeley National Laboratory) Guetersloh, S. (Lawrence Berkeley National Laboratory) Adams, J. (NASA Marshall Space Flight Center) Christl, M. (NASA Marshall Space Flight Center) Pinsky, Lawrence Ph.D. (University of Houston Physics Department) Sihver, Lembit Ph.D. (Chalmers University of Technology)		
Grant/Contract No.:	H-34854D		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This task calls for accelerator measurements of proton and heavy ion interactions in elemental targets and radiation shielding materials at energies characteristic of galactic cosmic radiation (GCR). Two types of measurements will be made: (1) fragmentation cross sections, or interaction probabilities in thin elemental targets and (2) fluences of fragments produced by interactions in thick elemental and composite targets.
The range of projectile atomic numbers for these measurements will be 1-26. The range of projectile energies will be 100-2000 MeV/nucleon. Differential cross sections for production of charged nuclear fragments, neutrons and (if deemed necessary) pions will be measured as a function of one or more observables including projectile energy, projectile mass, fragment charge, fragment energy and emission angle, according to the requirements defined by the model calculations. Projectiles, projectile energies, targets and other parameters will be selected in consultation with the developers of the transport models.

The detectors to be used will include individual and position-sensitive silicon detectors, silicon strip detectors and highly pixilated arrays. The latter two will be used to help separate out multiple fragments, thereby improving the accuracy of the cross sections for light nuclei. Liquid and solid plastic scintillators will be used to measure neutron energy spectra.

The accelerator facilities to be used will be the NASA Space Radiation Laboratory at Brookhaven National Laboratory), the National Institute of Radiological Sciences Heavy Ion Medical Accelerator at Chiba (HIMAC), and perhaps others, such as the proton synchrotron at the Loma Linda University Medical Center and the National Superconducting Cyclotron Laboratory. To improve the accuracy of nuclear transport models for space radiation applications. These models will be needed to optimize the design of future spacecraft with respect to protecting astronauts from galactic cosmic rays. Accurate modeling will reduce the necessity to physically test all candidate materials and designs, an expensive and time-consuming process. These measurements will be used to improve the cross section database used in these models both directly and through improvements in nuclear fragmentation models. The goal is obtain transport models that are accurate to ±25% for the fluxes of the major components emerging from the shielding. Comparing measured fluences with model calculations will measure this accuracy. Space radiation has been identified by the National Academy of Sciences as one of two potential “show stoppers” for the human exploration of space. Although high energy heavy ions comprise only a small fraction of the particles in space, their large energy deposition makes them a significant contributing factor to health risk for crewmembers on long-duration space missions. Interactions of the incident heavy ions in spacecraft or planetary habitat walls or in crewmembers’ bodies can modify an initially well-known radiation field. This complicates both spacecraft and habitat shielding design and evaluation of radiation risk.

It has been recognized for some time that understanding the effects of space radiation will require a combined experimental and modeling effort. Accelerator measurements have several advantages over those made in flight: at accelerators data can be gathered rapidly under controlled conditions; ground-based detectors have fewer constraints on size, power and complexity than do flight instruments; and while access to heavy ion accelerators is somewhat limited, it is much easier to obtain than access to space. On the other hand, it is not possible at an accelerator to exactly replicate the radiation found in space, operational spacecraft have complex internal structures that are difficult to replicate on the ground, and new candidate shielding materials are constantly being proposed.

Consequently, a great deal of effort has gone into modeling radiation transport in materials. Reducing the uncertainties in the model predictions is critical: the greater the uncertainty, the greater the weight penalty in the form of increased shielding thickness required under the ALARA (As Low As Reasonably Achievable) principle of radiation protection practice.

The GCR environment and the basic physics of the interactions of high energy charged particles in matter are well understood, but accurate and precise models require accurate and precise inputs and, for testing purposes, detailed characterization of the radiation produced by given incident radiation. Similar arguments apply in the case of the self-shielding of critical organs within the human body.

The energy range of greatest interest for space radiation applications is comparable to what has been available for over twenty-five years at heavy ion accelerators However, since accelerator experiments have been motivated for the most part by basic research in nuclear physics and astrophysics and by the needs of the charged particle radiotherapy community, the data are still limited in some areas of particular interest for space radiation. This proposal aims to remedy this.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Measurements:
HIMAC (Feb. 2005) BNL AGS (July 2005)

Cross-sections:

Analysis completed:

16O (290 and 400 MeV/u); 14N (290 MeV/u); 48Ti (1000 MeV/u)

Materials:

Analysis completed:

All NASA-LaRC targets exposed to 600 and 1000 MeV/u 16O All ORNL targets exposed to 600 and 1000 MeV/u 16O All University of Utah targets exposed to 600 and 1000 MeV/u 16O Processed NASA-LaRC targets exposed to 40 MeV protons at LBNL 88" Cyclotron. (Approx. 60 targets, total.)

Finalizing paper on polyethylene as a reference material for materials tests. This will include an in depth analysis of the performance of polyethylene under a variety of conditions with model comparisons.

Finalizing paper for publication on the performance of a variety of materials as radiation shields exposed to 1 GeV/u 56Fe

Neutrons:

Task Progress:	Analysis completed: 500 MeV/A Fe in stopping targets: neutron double-differential yields at 90, 120 and 150 degrees. 600 MeV/A Si + C, Cu, Pb: neutron double-differential cross sections between 5 and 80 degrees. (1) 600 MeV/A Ne + marsbar, 400 MeV/A Ne + ISS wall, and 290 C + marsbar, neutron double-differential cross sections. In press. (2) Neutron cross sections from 230 MeV/A He, 400 MeV/A N,400 MeV/A Kr, 400 MeV/A Xe, 500 MeV/A Fe, and 600 MeV/A Si. Paper written, to be sent to coauthors for review. Paper will be submitted to Nucl. Sci. and Eng. (3) Handbook on secondary particle production from heavy ion interactions. Publication date is January 2006. On-line database: Set up web site with secure login. Began work on formatting cross section data into HTML for web access. Presentations: Humans in Space conference, Graz, 5/05. (Guetersloh, Zeitlin) NASA Space Radiation Health Program, Port Jefferson, 5/05 (Miller, Zeitlin) IEEE Aerospace Conference, Big Sky, Montana, 3/05 (Miller)
Bibliography Type:	Description: (Last Updated: 01/21/2020)
Articles in Peer-reviewed Journals	La Tessa C, Guetersloh S, Heilbronn L, Miller J, Sihver L, Zeitlin C. "Fragmentation of 1 GeV/nucleon iron ions in thick targets relevant for space exploration." Adv Space Res. 2005;35(2):223-9. PubMed PMID: 15934198 ; https://doi.org/10.1016/j.asr.2005.02.007 , Jun-2005
Articles in Peer-reviewed Journals	Walker SA, Tweed J, Wilson JW, Cucinotta FA, Tripathi RK, Blattnig S, Zeitlin C, Heilbronn L, Miller J. "Validation of the HZETRN code for laboratory exposures with 1A GeV iron ions in several targets." Adv Space Res. 2005;35(2):202-7. PubMed PMID: 15934195 ; https://doi.org/10.1016/j.asr.2005.02.077 , Jun-2005
Books/Book Chapters	Nakamura T, Heilbronn LH. "Handbook on Secondary Particle Production and Transport by High-Energy Heavy Ions." World Scientific, 2005. https://doi.org/10.1142/5973 , Dec-2005